

# Aqueeb Anjum Sunny

akib.sunny@gmail.com | +8801923011378 | GitHub | Portfolio | Dhaka, Bangladesh

## Research Interests

Advanced reactors, Core design, Neutronics, Fuel performance optimization, ML in nuclear engineering

## Education

**Military Institute of Science and Technology (MIST)**

Jul 2025

Bachelor of Science in Nuclear Engineering

CGPA: 3.46 out of 4.00

## Research Experience

- Conducted Monte Carlo simulations to analyze the neutronic behavior of the ALFRED reactor core.
- Compared the neutronic performance and activation of Lead and LBE as LFR coolants.
- Evaluated reflector materials in an LFR core to assess neutron economy and fuel cycle length.
- Performed preliminary neutronic evaluation of a conceptual Multi-Purpose Research Reactor.
- Analyzed neutronics parameters of different annular fuels (UO<sub>2</sub>, UN, and UC<sub>2</sub>)

## Publications

1. Shuddho, S. S., **Sunny, A. A.**, & Mollah, A. S. (2025). *Neutronic Performance of Reflector Materials in Lead-Cooled Fast Reactor*. Nuclear Engineering and Design (Under Review). <https://dx.doi.org/10.2139/ssrn.5348419> Jul 2025
2. Dipto, R. R., Shuddho, S. S., **Sunny, A. A.**, & Mollah, A. S. (2024). *Analysis of Neutronics Parameters of Different Annular Fuel Using Monte Carlo Code OpenMC Utilizing JEFF-3.3 and ENDF/B-VIII.0 Nuclear Data Libraries*. Proceedings of the Energy Conference 2023. <https://dx.doi.org/10.2139/ssrn.4997514> Oct 2024

## Technical & Research Skills

**Reactor Physics Tools:** OpenMC, NJOY, ARMI

**Programming Languages:** Python (numpy, pandas, matplotlib, scikit-learn), C, MATLAB

**Version Control & Environment:** Git, Linux, Markdown

**Research Skills:** Core Design, Monte Carlo Simulations, Data Analysis, Nuclear Data Processing

## Professional Training

**Rooppur Nuclear Power Plant, Pabna**

7-8 Feb 2024

- Trained in nuclear power plant operations and safety protocols

**Bangladesh Atomic Energy Centre, Dhaka**

11-15 Feb 2024

- Completed Non-Destructive Testing training program

**TRIGA Research Reactor, Atomic Energy Research Establishment, Savar**

5 Mar 2024

- Studied research reactor operations and neutronics applications

## Courses & Workshops

Machine Learning Crash Course, Google

Sep 2025

2025 Nuclear Engineering Summer School, MTV

Aug 2025

Computational Nuclear Science and Engineering, IAEA

Jul 2025

Leadership & Outreach

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>MIST Nuclear Engineering Club, Senior Executive Panel</b>                                                                                                     | 2023 - 2024    |
| Organized nuclear engineering events, including a quiz competition for 50+ students, and delivered presentations to 30+ freshmen, introducing them to the field. |                |
| <b>Private Tutor (Physics &amp; Mathematics, High School Level)</b>                                                                                              | 2022 - Present |
| Provided individualized and group tutoring for higher-secondary students.                                                                                        |                |

Reference:

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**Prof. Dr. Abdus Sattar Mollah**  
Professor, Dept. of Nuclear Science & Engineering (NSE)  
Military Institute of Science and Technology (MIST)

**Relationship:** Thesis Supervisor  
**Email:** mosattar54@gmail.com  
**Phone:** +8801552316781  
**Address:** Dept. of NSE, MIST, Mirpur Cantonment, Dhaka-1216, Bangladesh.